

# Last Time

Populations vs. Samples (quick)

Statistics and Sample Spaces

Distributions

Quantiles

Distribution Shapes

Central Tendency

# Today

Properties of Central Tendency Measures

Summation (Sigma) Notation

Summation Rules

The Mean as a Geometric Center

Dispersion

Least Squares Principle

Skewness and Kurtosis

Outliers

Z-scores

# Properties of Central Tendency Measures

## Shoe Sizes

$X = \{$

- ▶ Mode
- ▶ Median
- ▶ Mean

$\}$

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  - ▶ Minimizes the Mean Absolute Deviation (MAD)
- ▶ Mean
  - ▶ Appropriate for interval or higher scales of measurement
  - ▶ Very sensitive to outliers
  - ▶ Is the unique centroid/geometric center/balance point of a distribution
  - ▶ Minimizes the Sum of Squared Deviations

# Sensitivity to Outliers

Robert Pershing Wadlow (1918-1940)  
(Let's go back!)



# Deviation Scores

$$D_i = X_i - \bar{X}$$

- ▶ What is  $\sum_{i=1}^N D_i$ ?
- ▶ What is  $\sum_{i=1}^N D_i^2$ ?



# Sum of Squares

$$\sum_{i=1}^N D_i^2 = \sum_{i=1}^N (X_i - \bar{X})^2$$

# Summation (Sigma) Notation

Example

$$\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i$$

# Summation Notation

$$\sum_{i=M}^N X_i = X_M + X_{M+1} + \cdots X_N$$

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$$(X_1 + X_2 + X_3 + X_4)/4 =$$

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Rewrite the following in summation notation:

$$X_1 Y_1 + X_2 Y_2 + \cdots + X_N Y_N =$$

# Summation Notation

$$\sum_{i=M}^N X_i = X_M + X_{M+1} + \cdots X_N$$

Rewrite the following in summation notation:

$$(X_1^2 - 2X_1 + 3) + 2(X_2^2 - 2X_2 + 3) + \cdots + N(X_N^2 - 2X_N + 3) =$$

# Summation Notation

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Rewrite the following in extended form:

$$\sum_{i=5}^7 X_i =$$

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$$\sum_{i=1}^6 X_i^2 =$$



# Summation Notation

$$\sum_{i=M}^N X_i = X_M + X_{M+1} + \cdots X_N$$

Rewrite the following in extended form:

$$\sum_{i=1}^5 \sum_{j=1}^4 (X_{ij}^2 + Y_i) =$$

# Summation Rules

- ▶ *Rule 1: The Sum of a Variable  $V$ .*

$$\sum_{i=1}^n V_i = V_1 + V_2 + \cdots + V_n.$$

- ▶ *Rule 2: The Sum of a Constant.*

$$\sum_{i=1}^n c$$

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- ▶ *Rule 2: The Sum of a Constant.*

$$\sum_{i=1}^n c = c + c + \cdots + c = nc.$$

# Summation Rules

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- *Rule 3: The Sum of a Product of a Constant and a Variable.*

$$\sum_{i=1}^n cV_i = cV_1 + cV_2 + \cdots + cV_n = c(V_1 + \cdots + V_n) = c \sum_{i=1}^n V_i, \text{ and}$$

$$\sum_{i=1}^n \frac{V_i}{c} = \frac{1}{c} \sum_{i=1}^n V_i,$$

# Summation Rules

- ▶ *Rule 4: Distribution of Summation.*

$$\sum_{i=1}^n (V_i + W_i) =$$

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$$\begin{aligned}\sum_{i=1}^n (V_i + W_i) &= (V_1 + W_1) + (V_2 + W_2) + \cdots + (V_n + W_n) \\ &= (V_1 + \cdots + V_n) + (W_1 + \cdots + W_n) \\ &= \sum_{i=1}^n V_i + \sum_{i=1}^n W_i,\end{aligned}$$



# Summation Rules

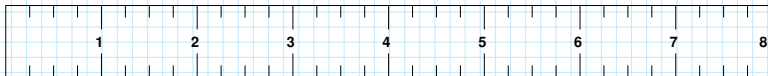
► *Rule 4: Distribution of Summation.*

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$$\sum_{i=1}^n (V_i - W_i) = \sum_{i=1}^n V_i - \sum_{i=1}^n W_i.$$

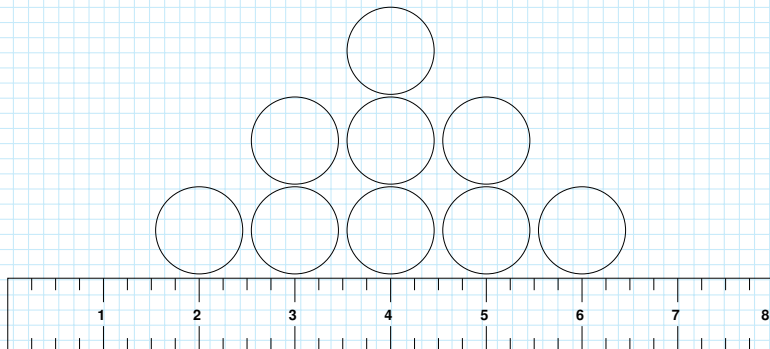
# The Balance Point

$\{2, 3, 3, 4, 4, 4, 5, 5, 6\}$



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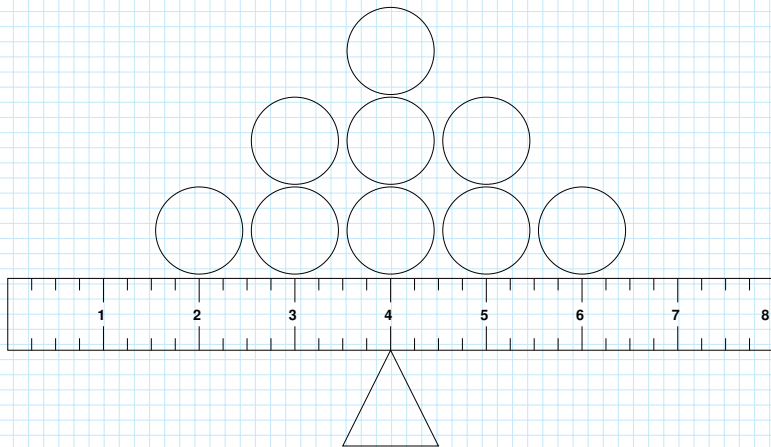
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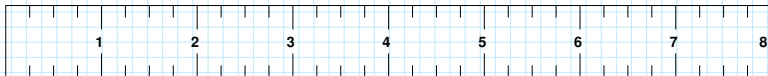
$\{2, 3, 3, 4, 4, 4, 5, 5, 6\}$

$$\bar{X} = \frac{1}{9} \sum_{i=1}^9 X_i = \frac{1}{9} (2 + 3 + 3 + \cdots + 6) = \frac{36}{9} = 4$$



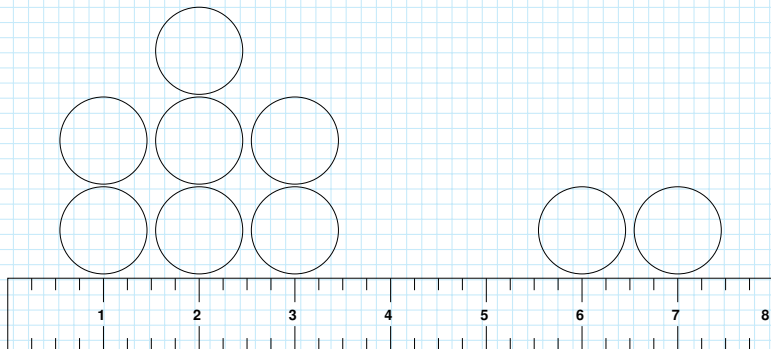
# The Balance Point

$\{1, 1, 2, 2, 2, 3, 3, 6, 7\}$



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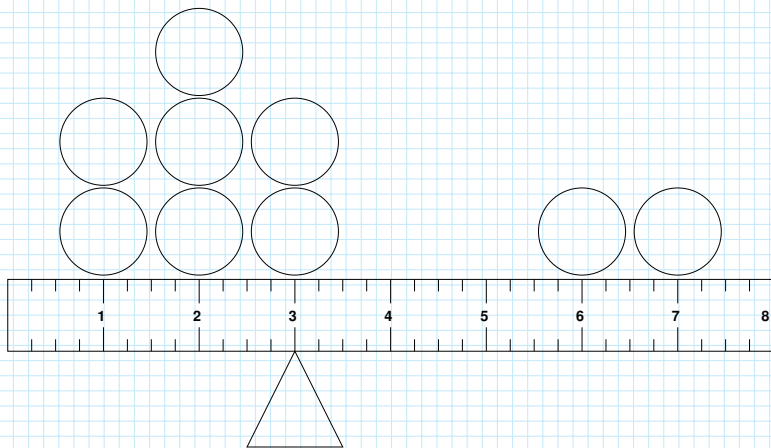
$\{1, 1, 2, 2, 2, 3, 3, 6, 7\}$



# The Balance Point

$\{1, 1, 2, 2, 2, 3, 3, 6, 7\}$

$$\bar{X} = \frac{1}{9} \sum_{i=1}^9 X_i = \frac{1}{9}(1 + 1 + 2 + \cdots + 7) = \frac{27}{9} = 3$$



## Proof that the Mean is the Balance Point

*Definition:* For a sample of scores  $\{X_1, X_2, \dots, X_N\}$ , the *deviation scores*  $\{D_1, D_2, \dots, D_N\}$  are all the differences between the scores ( $X$ ) and their mean ( $\bar{X}$ ).

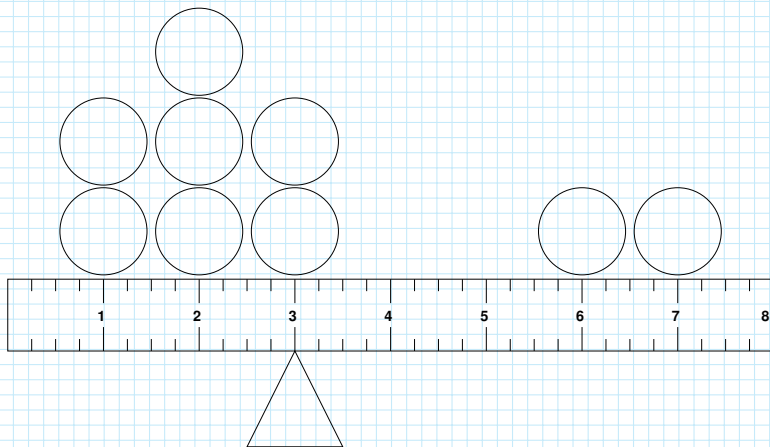
$$D_i = X_i - \bar{X}$$



## Proof that the Mean is the Balance Point

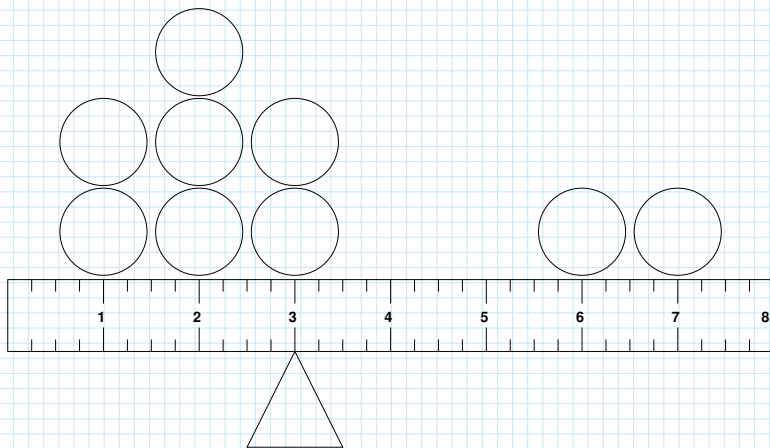
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$$D_i = X_i - \bar{X}$$



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For the mean to “balance” the scores, for every  $D_i < 0$  there must be some positive  $D_j$ s > 0 that zero it out.



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For the mean to “balance” the scores, for every  $D_i < 0$  there must be some  $D_j > 0$  that zero it out. That is,

$$\sum_{i=1}^N D_i = 0.$$

# Proof

$$\sum_{i=1}^N D_i = \sum_{i=1}^N (X_i - \bar{X})$$

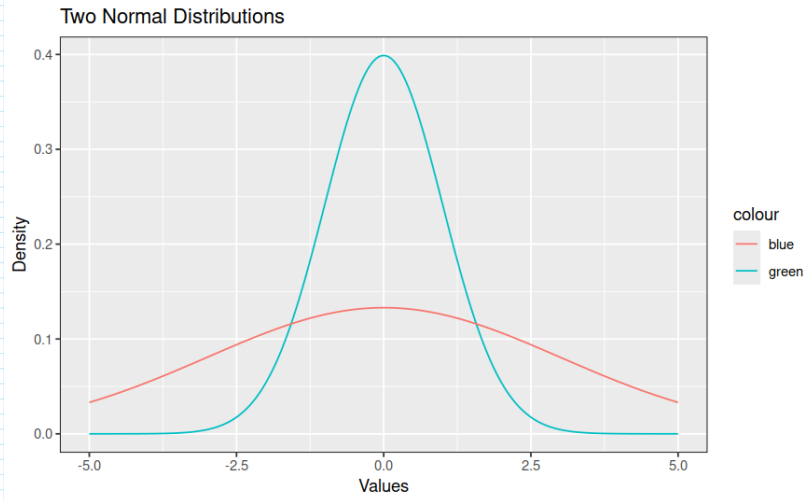
## Proof

$$\begin{aligned}\sum_{i=1}^N D_i &= \sum_{i=1}^N (X_i - \bar{X}) \\&= \sum_{i=1}^N X_i - \sum_{i=1}^N \bar{X} \\&= \sum_{i=1}^N X_i - N\bar{X} \\&= \sum_{i=1}^N X_i - N \frac{1}{N} \sum_{i=1}^N X_i \\&= \sum_{i=1}^N X_i - \sum_{i=1}^N X_i = 0.\end{aligned}$$

# Dispersion

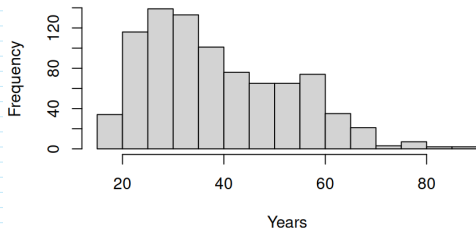
- ▶ By how much do the values in the sample deviate from the “most characteristic score”?
- ▶ The central tendency is a prediction about the other values in the sample.

# Dispersion

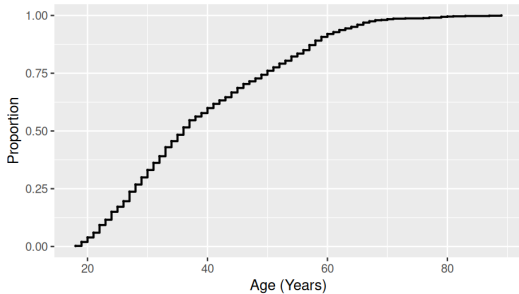


# Range

**Respondent Ages**



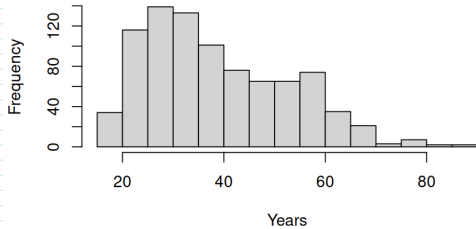
**Cumulative Relative Frequency Distribution**



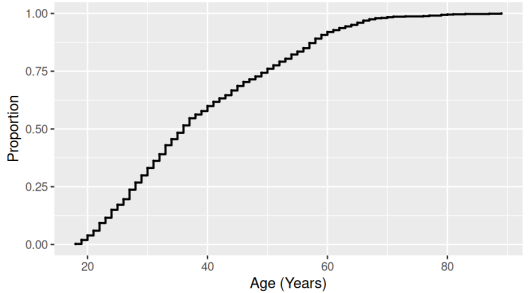


# Interquartile and Semi-interquartile Range

**Respondent Ages**



**Cumulative Relative Frequency Distribution**



# Variance and Standard Deviation

$$D_i = X_i - \bar{X}$$

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$$s^2 = \frac{\sum_{i=1}^N D_i^2}{N} = \frac{\sum_{i=1}^N (X_i - \bar{X})^2}{N}$$

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$$S = \sqrt{S^2} = \sqrt{\frac{\sum_{i=1}^N (X_i - \bar{X})^2}{N}}$$

## Population vs. Sample Variation

$$S^2 = \frac{\sum_{i=1}^N D_i^2}{N} = \frac{\sum_{i=1}^N (X_i - \bar{X})^2}{N}$$

$$s^2 = \frac{\sum_{i=1}^N D_i^2}{N - 1} = \frac{\sum_{i=1}^N (X_i - \bar{X})^2}{N - 1}$$

# Properties

- ▶ Range
  - ▶ Simple to compute
  - ▶ Unstable, very sensitive to outliers
- ▶ Interquartile Range
  - ▶ Less easy to compute
  - ▶ Robust
- ▶ Variance
  - ▶ Easy to compute
  - ▶ Sensitive to outliers
  - ▶ Unintuitive
- ▶ Standard deviation
  - ▶ Easy to compute
  - ▶ Intuitive

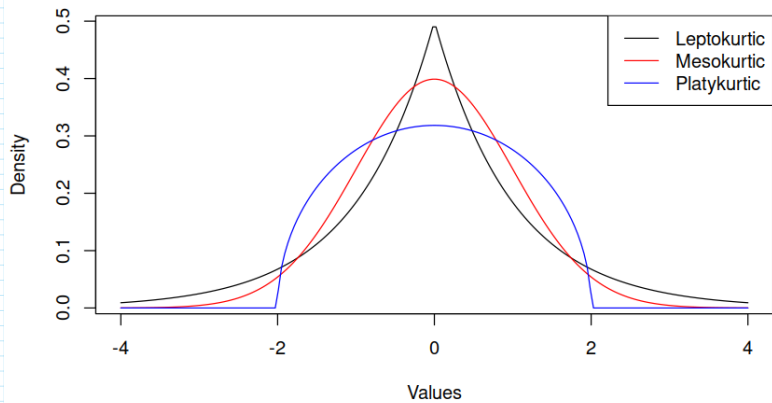
# Least Squares

- ▶ How well does a measure of central tendency  $C$  characterize all the values in a sample?
- ▶ Sum of squared deviations is a loss function  $L(C)$  around the center  $C$ .

$\bar{X}$  is a least-squares estimator

$$L(C) = \sum_{i=1}^N (X_i - C)^2$$





# Skewness and Kurtosis

$$Sk = \frac{\sum_{i=1}^N (X_i - \bar{X})^3 / N}{S^3}$$

$$Kur = \frac{\sum_{i=1}^N (X_i - \bar{X})^4 / N}{S^4} - 3$$

# Outliers

- ▶ Chebyshev's Inequality:

$$Pr(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

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- ▶ Chebyshev's Inequality:

$$Pr(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

- ▶ An observation  $X$  can't get very far from its population mean  $\mu$ .

## A Potential Strategy

- ▶ Pick a value for  $k$ , say  $k = 4$ , so  $k^2 = 16$  and  $1/k^2 = 0.06$
- ▶ Compute  $\bar{X}$  and  $s^2$  from the sample
- ▶ Suspect any value outside of  $\bar{X} \pm 4s$  might be an outlier because the probability of being that far away is very small

# Z-scores

- ▶ A location-scale transformation of values  $X$
- ▶ “Standard” scores with mean  $\bar{Z} = 0$  and variance  $s_Z^2 = 1$
- ▶ Does NOT make scores “normally” distributed!
- ▶ Describes a score's position relative to the mean: how many standard deviations is it from the mean?

$$Z_i = \frac{D_i}{s} = \frac{X_i - \bar{X}}{s}$$